

Himanshu Sharma

Postdoctoral researcher in experimental high-energy nuclear physics

INFN Fellow and member of the ALICE Collaboration at CERN.

Istituto Nazionale di Fisica Nucleare (INFN), Section of Padova, Italy.

Profile

Postdoctoral researcher in experimental high-energy nuclear physics (ALICE Collaboration at CERN) specializing in charmonium production, heavy-flavor hadronization, and machine learning applications. Proven track record in leading complex data analyses, calibrating detector simulations, and serving on international collaboration review committees. Recognized for providing high-precision measurements that test and constrain QCD models, supported by an active portfolio in university-level lecturing, student mentorship, and science communication.

Keywords Strong interactions, quantum chromodynamics (QCD), quark–gluon plasma (QGP), high-energy nuclear collisions, charmonium and heavy-flavor production, jet substructure, machine learning applications.

Research and education

Feb 2023 –
present

Postdoctoral researcher, ALICE experiment

Istituto Nazionale di Fisica Nucleare (INFN), Padova, Italy

The contract began on 6 February 2023, during the final thesis-writing phase of the PhD. The doctoral degree was awarded on 4 December 2023.

Physics measurements

Charm-strange baryon (Ξ_c^+) production in pp and Pb–Pb collisions in LHC Run 3 (2022–26), using machine learning techniques.

Machine learning-based signal selection. Extracted the invariant-mass signals of $\Xi_c^+ \rightarrow pK^-\pi^+$ and $\Lambda_c^+ \rightarrow pK^-\pi^+$ with Boosted Decision Trees (BDTs) to optimize signal-to-background separation, particularly in the low-transverse-momentum region, in pp collisions.

Physics analysis and dissemination. Measured the charm baryon-to-baryon ratio Ξ_c^+/Λ_c^+ in pp collisions at $\sqrt{s} = 13.6$ TeV, compared the results with theoretical QCD models, and presented these findings for the first time in two talks (Talk 1, Talk 2) at the SQM 2026 conference.

Software development

Software tools for data-driven calibration of Monte Carlo simulations.

Main developer of the software implementation of a method that modifies the spatial and momentum resolutions of reconstructed tracks in the simulation to reproduce those estimated in data with data-driven techniques. The approach has since been adopted in multiple ALICE heavy-flavor analyses to help reduce systematic uncertainties. This is crucial for the measurements of all hadrons decaying far from the primary vertex, especially hadrons containing charm or beauty quarks.

Oct 2018 –
Dec 2023

Doctoral researcher, ALICE experiment

Institute of Nuclear Physics, Polish Academy of Sciences (IFJ PAN), Kraków, Poland

PhD in high-energy nuclear physics. Advisor: Dr. hab. Jacek Otwinowski. Thesis defended on 24 November 2023, degree awarded on 4 December 2023.

Thesis: “Prompt and non-prompt J/ψ production in Pb–Pb collisions at $\sqrt{s_{NN}} = 5.02$ TeV with ALICE experiment”

Physics measurements

Prompt and non-prompt J/ψ production in Pb–Pb collisions at $\sqrt{s_{NN}} = 5.02$ TeV in the ALICE experiment.

Extraction of non-prompt J/ψ fractions using two-dimensional simultaneous likelihood fits on the invariant mass and decay length of J/ψ candidates, reconstructed via the $J/\psi \rightarrow e^+e^-$ decay channel at midrapidity using datasets collected during LHC Run 2.

Determination of nuclear modification factors and production yields of prompt and non-prompt J/ψ as a function of transverse momentum and collision centrality.

Detector development

Contribution to the development of the ALICE Fast Interaction Trigger (FIT) detector for LHC Run 3 during Long Shutdown 2 (LS2).

Tested the signal and high-voltage setup for all channels of the FIT-FT0 detector. Also contributed to detector commissioning, taking part in cosmic-ray data taking and data analysis.

Tested the performance of FIT-FV0 detector channels with laser signals after commissioning at the LHC, CERN (Link 1, Link 2).

FIT software: developed time-calibration tasks in the Quality Control framework for the FV0 detector (PR), and tested time calibration for FV0 channels with laser and simulation data (Link).

Jul –
Oct 2018

Research assistant

Indian Institute of Technology (IIT) Indore, India

Investigated freeze-out temperatures and radial flow in proton–proton collisions at $\sqrt{s} = 7$ TeV by analyzing the transverse-momentum spectra of produced charged and strange hadrons.

Acquired expertise in thermodynamic modeling (Blast-Wave models and Tsallis statistics) to extract key thermodynamic parameters and describe the collective expansion of the system.

Aug 2016 –
Jul 2018

M.Sc. Physics

Indian Institute of Technology (IIT) Madras, India

Thesis: “Elliptic flow of charged hadrons in Pb–Pb collisions at $\sqrt{s_{NN}} = 2.76$ TeV using the AMPT”. Advisor: Dr. Prabhat Pujahari.

Studied the collective behavior of the hot, dense medium produced in Pb–Pb collisions at $\sqrt{s_{NN}} = 2.76$ TeV by analyzing azimuthal anisotropy observables using the AMPT event generator.

Acquired expertise in heavy-ion event generators, relativistic kinematics, and heavy-ion phenomenology, with a focus on characterizing elliptic flow as a smoking-gun signature of QGP formation.

Aug 2012 –
Aug 2015

B.Sc.

University of Kota, India

Publications

As a member of the ALICE Collaboration I am a co-author of more than 250 publications. The complete list is on my INSPIRE-HEP profile.

Journal articles

- | | |
|------|---|
| 2024 | Prompt and non-prompt J/ψ production at midrapidity in Pb–Pb collisions at $\sqrt{s_{NN}} = 5.02$ TeV ALICE Collaboration, JHEP 02 (2024) 066. Main analyzer and paper committee member. |
| 2019 | Radial flow and differential freeze-out in pp collisions at the LHC at $\sqrt{s} = 7$ TeV Eur. Phys. J. A 55, 3 (2019). Co-author and analyzer. |

Conference proceedings

Written for the ALICE Collaboration, unless noted.

2026	Studying charm-quark hadronization via charm-baryon production measurements in pp at the LHC with ALICE EPJ Web Conf. 364 (2026) 08010
2025	Investigation of early magnetic field and angular momentum in ultrarelativistic heavy-ion collisions via D^{*+} spin alignment with ALICE EPJ Web Conf. 316 (2025)
2023	Recent charmonium measurements in Pb–Pb collisions with ALICE PoS EPS-HEP (2023) 211
2022	Prompt and non-prompt J/ψ production at midrapidity in Pb–Pb collisions at $\sqrt{s_{NN}} = 5.02$ TeV with ALICE PoS LHCP (2022) 339
2022	Charmonium as a probe of hot quark matter in nuclear collisions with ALICE at the LHC Acta Phys. Pol. B Proc. Suppl. 15, 3-A37 (2022)
2022	Non-prompt J/ψ measurements at midrapidity in pp, p–Pb and Pb–Pb collisions with ALICE PoS PANIC (2022) 239
2018	An overview of differential freeze-out and radial flow in pp collisions at $\sqrt{s} = 7$ TeV at the LHC DAE Symp. Nucl. Phys. 63 (2018) 982–983. Co-author and analyzer.

Conference presentations

Parallel talks

2026	Testing pQCD calculations and hadronisation with charm and beauty hadrons in pp collisions with ALICE International Conference on High Energy Physics (ICHEP) 2026, Natal, Brazil, 31 Jul – 5 Aug 2026
2025	Studying charm-quark hadronization via charm-baryon production measurements in pp at the LHC with ALICE Quark Matter (QM) 2025, Frankfurt, Germany, 6–12 April 2025
2024	D^{*+} spin alignment in ultrarelativistic heavy ion collisions in ALICE Strangeness in Quark Matter (SQM) 2024, Strasbourg, France, 3–7 May 2024
2023	Recent charmonium measurements in Pb–Pb collisions with ALICE European Physical Society High Energy Physics (EPS-HEP) Conference 2023, Hamburg, Germany, 21–25 Aug 2023

2022	Prompt, non-prompt, and coherent J/ψ production in Pb–Pb collisions with ALICE EMMI Conference 2022, Kraków, Poland, 19–22 Sept 2022
2022	Inclusive, prompt, and non-prompt J/ψ production at midrapidity in Pb–Pb collisions at 5.02 TeV with ALICE International Conference on High Energy Physics (ICHEP) 2022, Bologna, Italy, 6–13 July 2022
2022	Charmonium as a probe of hot quark matter in nuclear collisions in ALICE at the LHC Epiphany 2022, Kraków, Poland, 10–14 Jan 2022 (online), student session
2021	Beauty production in pp and Pb–Pb collisions with ALICE MESON 2021, 17–21 May 2021 (online)

Posters

2022	Prompt and non-prompt J/ψ production at midrapidity in Pb–Pb collisions at 5.02 TeV with ALICE Large Hadron Collider Physics Conference (LHCP) 2022, 16–20 May 2022 (online)
2022	Nuclear modification factor of prompt and non-prompt J/ψ in Pb–Pb collisions at 5.02 TeV at midrapidity with ALICE Quark Matter (QM) 2022, Kraków, Poland, 4–10 April 2022
2021	Non-prompt J/ψ measurements at midrapidity in pp, p–Pb and Pb–Pb collisions with ALICE Particles and Nuclei Conference (PANIC) 2021, 5–10 Sept 2021 (online)
2021	Measurements of non-prompt J/ψ -meson at midrapidity in Pb–Pb collisions with ALICE at the LHC Initial Stages (IS) Conference 2021, 10–15 January 2021 (online)

Roles in the ALICE Collaboration

Internal and analysis review committees

Reviewing ALICE paper drafts, checking the consistency of the analysis procedure, and suggesting improvements to the quality of the measurements.

IRC member	Multiplicity-dependent Ξ_c^+ and Ξ_c^0 production in pp collisions at $\sqrt{s} = 13$ TeV (submitted to JHEP)
ARC member	Measurement of D^{*+} spin alignment in pp collisions at $\sqrt{s} = 13.6$ TeV
ARC member	Double charm production measurements: D^0 – J/ψ associated production and D^0 – D^0 production in pp collisions at $\sqrt{s} = 13.6$ TeV
ARC member	Lifetime measurements of Λ_c^+ and Ω_c^0 baryons in pp collisions at $\sqrt{s} = 13.6$ TeV

Detector operations and data taking

Mar – Apr 2024	Shift Leader and Detector Control System (DCS) shifter. Managed real-time detector operations, safety protocols, and data-taking quality control during active LHC runs.
Jul 2023	Data Quality Control (QC) shifter. Monitored live data streams to identify, troubleshoot, and document detector anomalies, ensuring high-quality data preservation.

Data quality and software contributions

Feb – Dec 2023	Global track quality monitoring in reconstructed data. Systematic evaluation of global track reconstruction quality in reconstructed datasets from pp and Pb–Pb collisions. Processed large-scale data samples and presented weekly quality control assessments at ALICE QA collaboration meetings.
Jan – Dec 2024	Track impact parameter performance. Evaluated and monitored track impact parameter performance across LHC Run 3 datasets (pp and Pb–Pb collisions). Conducted comparative studies between data and anchored Monte Carlo simulations, presented regularly in internal ALICE working groups.
Ongoing	Monte Carlo simulation tuning (ALICE Padova group). Developed and implemented data-driven parametrizations to tune impact parameter distributions in Monte Carlo simulations, significantly improving simulation-to-data agreement for the collaboration.

Funding and awards

Fellowships and grants

2025 – 2027	INFN Postdoctoral Research Fellowship , INFN Section of Padova, Italy (2 years, total 77,447 EUR)
2023 – 2025	Postdoctoral Research Grant (INFN & STRONG-2020) , INFN Padova, Italy (24 months)
2023	Postdoctoral Research Grant , University of Padova, Italy (4 months, total 5,600 EUR)
2023	CNRS Postdoctoral Fellowship , IJCLab, University of Paris-Saclay, France (declined)
2018 – 2022	Doctoral Fellowship , Institute of Nuclear Physics, Polish Academy of Sciences (IFJ PAN)
Jan 2022	NAWA Student Exchange Grant , University and INFN Section of Cagliari, Italy (2-week scientific visit, 10,000 PLN / ~2,500 EUR)

Awards and qualifications

2024	INFN Fellowship Award, ranked 3rd globally (score 95.7%). Awarded the Senior Level 3 Research Grant in Experimental Physics. Only 20 recipients are selected annually worldwide across experimental particle, nuclear, astroparticle physics, and technological research.
2016	Top 1% national rank, IIT-JAM. Secured All India Rank 195 out of thousands of candidates in the highly competitive national entrance examination for the M.Sc. in Physics.
2016, 2018	National research eligibility qualifications (India). Qualified in the Joint Entrance Screening Test (JEST-2016) and the Graduate Aptitude Test in Engineering (GATE-2018).

Teaching and mentoring

Lecturer	Erasmus Mundus Joint Master's degree program (NucPhys at UniPd): lectures on "Ultrarelativistic heavy-ion collisions" at the University of Padova. Designed the hands-on sessions for D^0-meson yield extraction. Designed and delivered a novel hands-on lecture on machine learning in heavy-ion collisions.
2021	Co-supervisor. Co-supervised two undergraduate summer students in the IFJ PAN Particle Physics Summer Student Programme (PPSS 2021), in a project on the application of machine learning to J/ψ signal extraction using electron–positron pairs in Pb–Pb collisions in ALICE.
2019	Summer school. Selected to attend Machine Learning in High-Energy Physics (MLHEP-2019), DESY, Hamburg, Germany.
Jan 2022	Research visit. 15-day internship at the University and INFN Cagliari, Italy, working on prompt and non-prompt J/ψ measurements in Pb–Pb collisions with Dr. Fiorella Fionda (Physics Working Group convener in ALICE).

Writing and outreach

Science communication on Medium, on physics, mathematics, and artificial intelligence.

Mar 2025	Prompt-less AI: A quest of motive
Oct 2025	How long do we care? The Nobel Prize and the attention span
Nov 2025	Irrational numbers on canvas: turning π and e into art
	Hand-picked and "Boosted" by the Medium curator team: distributed across 11,000+ feeds, with 500+ views and 200+ reads.

Public outreach

2018 – 2022

Exhibitor, Małopolska Researchers' Night, Kraków, Poland. Every September for four years. Designed and built a portable DIY cloud chamber for real-time tracking and demonstration of cosmic radiation to the general public (experimental tracking video).

Professional skills

Programming

C, C++, Python (including data visualization), bash scripting

Machine learning

Boosted Decision Trees (BDTs), multilayer perceptrons and neural networks, deep learning applications

Simulation

Heavy-ion simulations with the AMPT event generator

Analysis and tools

HEP data analysis with ROOT and RooFit, GitHub, LaTeX

Contact

Himanshu Sharma, PhD

Istituto Nazionale di Fisica Nucleare (INFN), Section of Padova, Italy

himanshu.sharma@cern.ch s.hymanshu@gmail.com

Last updated September 2026.